Analysis Accuracy of XGBoost Model for Multiclass Classification - A Case Study of Applicant Level Risk Prediction for Life Insurance

Widya Fajar Mustika
Department of Mathematics
Universitas Indonesia
Depok, Indonesia
widya.fm@sci.ui.ac.id

Hendri Murfi
Department of Mathematics
Universitas Indonesia
Depok, Indonesia
hendri@sci.ui.ac.id

Yekti Widyaningsih
Department of Mathematics
Universitas Indonesia
Depok, Indonesia
yekti@sci.ui.ac.id

***Abstract* - Risk level assessment for insurance applicants is an important part of life insurance, so it needs to be classified. Determination of the level of risk claims on life insurance is based on the applicant's historical data. Submission to become a member of a life insurance requires a short time. But the application of a machine learning model can help classify prospective insurance applicants based on the level of risk quickly. One machine learning model is Extreme Gradient tree boosting (XGBoost) which is a decision tree based model. This model is used to predict risk in life insurance. The missing values in the data used are overcome by several strategies in the data processing process to increase the accuracy value of the XGBoost model. The results of this study show that the accuracy of the XGBoost model is 0.60730 with kappa units which indicates that the XGBoost model is very good and can be applied to the problem of predicting the level of risk claims for life insurance applicants. When compared to the decision tree, random forest and Bayesian ridge models, the performance of the XGoost model still excels in processing missing values in the data used.**

Keywords— risk level prediction, big data, machine learning, multi-class classification, ensemble learning, XGBoost

I. INTRODUCTION

Insurance is an agreement between two parties, namely the insurer and insurance applicant (the insured) which is the basis for receiving premiums by insurance companies to provide compensation due to damage or loss due to an uncertain event [1]. Insurance is a risk transfer activity from one party to another party in which there are rules and principles compiled by both parties.

There are several types of insurance offered by insurance companies, one of which is life insurance. Life insurance is insurance that protects both individuals and families from unexpected financial losses, which are caused by death, lifelong disability, or no longer productive [2].

Risk is defined as the results or consequences caused by events that occur or that will occur in the future. The community needs life protection for a more prosperous life. In fighting for prosperity, it cannot be separated from the risks that arise. Risk prediction is one of the most effective ways to overcome financial shortages [3]. The risk in life insurance is caused by several factors, namely accidents, medical conditions or health, lifestyle, and age groups. The risks associated with life insurance claims are generally different for each person. Therefore, it is very important for life insurance companies to predict the level of risk for prospective insurance policyholders to determine the premium value according to the level of risk.

The applicant's risk level usually is categorized following the filing data held by the applicant. The manual categorization requires a lot of time because there are many life insurance applicants with various data, i.e., age, height, weight, work history, insurance history, medical history, family history [4]. Moreover, the historical data related to this need is also big. Machine learning is a common method to imitate human capability in fast predicting the level of risk [5]. From the machine learning point of view, the problem of the risk level prediction is supervised learning, that is, a multi-class classification problem.

XGBoost is a recent machine learning model that is popular because of its good accuracy and efficient out-of-core ensemble learning [6]. XGBoost uses a decision tree as a base learner and a generalization of the gradient boosting method to construct the next base learners. This model construction is rather different from another ensemble learning, e.g., random forest, where the construction a base learning does not depend on other base learners. Moreover, XGBoost is the sparsity aware algorithm that can handle the missing values of the historical data. It means that historical data containing missing values can be processed without any imputation preprocessing.

In this paper, we apply and analyze the accuracy of XGBoost for the problem of risk level prediction in life insurance. We compare the accuracy with ones of the decision tree and random forest with standard imputation

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preprocessing. Our simulations show that the XGBoost model outperforms both decision tree and random forest.

The rest of the paper is organized as follows: In section 2, the reviews of related works are presented. Section 3 describes the methods of research. Section 4 describes the implementation, simulation., result, and discussion. Finally, we give a conclusion in Section 5.

II. RELATED WORK

Big data is related to large data. Definition of big data is a term that describes a large volume of data at high speed, complex, and contains variables that require advanced techniques and technology to enable the capture, storage, distribution, management and analysis of information [7]. In Galicia's paper, it represents an ensemble model for forecasting large data time series. The ensemble model used consists of three methods (decision tree, gradient tree boosting, and random forest) which are proposed because the three results have been achieved in the big data application where the weights of the ensemble model are calculated using the least square method [8].

The use of the XGBoost model related to problems in the insurance industry, namely Chen and Guestrin (2016) has conducted research related to the comparison of the scalability of the XGBoost model and other models in the case of regression problems insurance of motor vehicle [6]. Fauzan and Murfi (2017) have implemented the XGBoost model in predicting whether policyholders "submit claims" or "not submit claims" on insurance claims [9]. Chang (2018) applied the XGBoost model in the construction of a financial credit risk assessment model. These three studies applied the XGBoost model in dealing with problems in life insurance, but cases in all three models handled cases of binary regression and classification [10]. Based on the related research, in this thesis will be applied and analyzed the extreme gradient boosting (XGBoost) model in the case of the multiclass classification to predict the level of risk in Prudential life insurance. The accuracy of the XGBoost model will be compared with several other machine learning models to see the performance of the XGBoost model.

III. METHODS

In this section, we discuss the methods used. The method used is the CART method, which is the basic method for the XGBoost model.

A. Classification and Regression Trees

The Classification and Regression Trees (CART) method was introduced by Leo Breiman in 1984. CART is a method of a decision tree that is included in nonparametric statistics [11]. CART is a binary decision tree that is built by dividing a node into two root vertices repeatedly. The basic idea of the CART tree formation is to choose to split between all possible splitting at each node so that the root node of the child produced is the purest.

For example, given training data, namely $\boldsymbol{D}$ as follows:

$$\boldsymbol{D} = \{(\boldsymbol{x}_i, y_i)\}, 1 \leq i \leq n \tag{1}$$

$$\boldsymbol{A} = [\boldsymbol{x}_1 | \boldsymbol{x}_2 | \dots | \boldsymbol{x}_n] \subset \mathbb{R}^{nm} \tag{2}$$

$$\boldsymbol{x}_i = [a_{ij} \in \boldsymbol{A}], 1 \leq j \leq m \tag{3}$$

$$\boldsymbol{v}_j = [a_{ij} \in \boldsymbol{A}],\ 1 \leq i \leq n \tag{4}$$

Where n is the number of life insurance applicants, m is the number of predictor variables in the data, $\boldsymbol{x}_i$ is the data for applicants for i applicant, $\boldsymbol{v}_j$ is all values of the j predictor variable, and y_i is the actual target, namely the risk level of life insurance applicants.

In the case of the multiclass classification, CART was built based on training data, so that training data with the same level or value would be grouped. The CART formation process is done by selecting features and threshold values or splitting rules that make the purest data partitions, observations with the same level or value grouped on each branch of the tree formed. In the case of multilevel classification, the output function f of CART is a structure of binary tree graphs with a level that corresponds to input data where each node of the graph contains a data set of life insurance applicants, so the function f in a data x, can be written

$$f(x) = w_{q(x)},\ q\colon \mathbb{R}^m \to \{1,2,\dots,T\},\ w \in \mathbb{R}^T \tag{5}$$

$$I_t = \{i | x_i \in t\}, 1 \leq t \leq T \tag{6}$$

$$w_t = \frac{|\{i \in I_t | y_i = 1\}|}{I_t}, 1 \leq t \leq T \tag{7}$$

where T is the number of leaf node at f,36 $q(\boldsymbol{x})$ is a function that maps data $\boldsymbol{x}$ to one of the leaf node at f, I_t index data which is on a leaf node t, and w_t, $1 \leq t \leq T$ weight of leaf on f. The purity of a node can be calculated with Gini impurity, which is defined by:

$$G = 1 - \sum_{i=1}^{K} (p_i)^2 \tag{8}$$

where K is many levels, which have probability p_i, with $i = 1, 2, \dots, K$. The value of this root node will be minimum if the data on the node only consists of one level or if the node contains data with a similar or homogeneous level.

B. Extreme Gradient Boosting (XGBoost)

The ensemble method consists of two main approaches, namely bagging and boosting. Bagging method, which is to build several models independently, is used to increase the accuracy of predictions sequentially at a given time, where the final prediction is the average prediction [12]. Boosting, which is to build several models in a sequence where the error function used to build a model depends on the performance of the previous model. The process of boosting method depends on the accuracy of the basic model of the weak hypothesis that is the smallest [13].

An example of a boosting approach is the gradient boosting method that can be implemented in various problems, both regression, and classification. Research on gradient boosting has been carried out from various

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sciences since the 1990s. Gradient boosting was first introduced by Leo Breiman (1984) that boosting can be interpreted as an appropriate optimization algorithm [14]. The boosting process is to minimize Root Mean Square Error (RMSE) sequentially, where the concept of gradient boosting lies in its development, namely the expansion of additional criterion fittings [15].

The Extreme Gradient Boosting (XGBoost) method is the development of a gradient boosting model that has superior results and process speeds. XGBoost has the advantage of processing missing value data without imputation of data at the beginning of learning. [16].

The XGBoost method applies the process of classification and regression trees (CART) as weak learners and increases the performance of each tree formed by minimizing its objective regularization function. The XGBoost method has good ability to predict, especially in the case of classification, because this method has concepts such as sparsity awareness algorithms for each tree that is formed as well as out of core calculations that aim to speed up computing [6].

Algorithm 1: Exact Greedy Algorithm for Split Finding

Input: I, instance set of the current node
Input: d, feature dimension
$gain \leftarrow 0$
$G \leftarrow \sum_{i \in I} g_i$, dan $H \leftarrow \sum_{i \in I} h_i$
for $k = 1$ to m:
  $G_L \leftarrow 0,\ H_L \leftarrow 0$
  for j in sorted do:
    $G_L \leftarrow G_L + g_j,\ H_L \leftarrow H_L + h_j$
    $G_R \leftarrow G - G_L,\ H_R \leftarrow H - H_L$
    $score \leftarrow \max[score, \frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}]$
  end
end
Output: Split with max score

Fig.1 Exact Greedy Algorithm for Split Finding

C. Sparsity Aware Split Finding

Missing values is information or data that is not available for an object given. Missing data because there is no data or information for an object, not found or indeed data or information is difficult to find. The existence of missing values with only a small number, for example, only 1% of all data, is basically not a problem for the entire dataset. However, if the percentage of the missing values is quite large, it is necessary to handle whether the data containing many of the missing is still feasible to be processed or not.

The novelty introduced by XGBoost is the ability to overcome missing values. When a value is a missing value in the dataset, it will be immediately classified using the default direction. The default direction of each node on the CART is formed through the sparsity aware split algorithm. The algorithm stage begins with the formation of a root node containing a subset of observation data, then the algorithm collects all observational data with the value of each feature not a missing value in the dataset I. Next, calculate the gain obtained from splitting to the right and left to each observation on dataset I. is then stored based on the maximum split direction for all observations on dataset I as the optimal value of features. This stage continues to repeat for each feature, and the default direction is obtained by taking the optimal value for each feature. Following are the sparsity algorithms aware of split finding [6],

Algorithm 2: Sparsity aware Split Finding

Input: $\boldsymbol{I}$, instance set of the current node
Input: $\boldsymbol{I_k = \{i \in I | x_{ik} \neq missing\ value\}}$
Input: $\boldsymbol{d}$, feature dimension
Also applies to the approximate setting, only collect statistic of non-missing entries into buckets
$\boldsymbol{gain \leftarrow 0}$
$\boldsymbol{G \leftarrow \sum_{i \in I} g_i,\ \ H \leftarrow \sum_{i \in I} h_i}$
for $\boldsymbol{k = 1}$ to $\boldsymbol{m}$ do
  // enumerate missing values goto right
  $\boldsymbol{G_L \leftarrow 0,\ \ H_L \leftarrow 0}$
  for $\boldsymbol{j}$ in sorted do
$\boldsymbol{G_L \leftarrow G_L + g_j,\ \ H_L \leftarrow H_L + h_j}$
    $\boldsymbol{G_R \leftarrow G - G_L,\ \ H_R \leftarrow H - H_L}$
$\boldsymbol{score \leftarrow \max\left(score, \frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}\right)}$
end
// enumerate missing values goto left
  $\boldsymbol{G_R \leftarrow 0,\ \ H_R \leftarrow 0}$
  for $\boldsymbol{j}$ in sorted do
    $\boldsymbol{G_R \leftarrow G_R + g_j,\ \ H_R \leftarrow H_R + h_j}$
    $\boldsymbol{G_L \leftarrow G - G_R,\ \ H_L \leftarrow H - H_R}$
$\boldsymbol{score \leftarrow \max\left(score, \frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{G^2}{H+\lambda}\right)}$
  end
end
Output: Split and default direction with max gain

Fig 2. Sparsity Aware Split Finding Algorithm

IV. IMPLEMENTATION, SIMULATION, RESULT, AND DISCUSSION

This section explains the research process, starting from the stages of data collection. Then processing the data to get the accuracy of the XGBoost model.

A. Data Sets

The data used in this study is risk level data from the Prudential Life Insurance company. This data shows information on life insurance applicants used to build and evaluate risk predictors. This data is obtained from the https://www.kaggle.com/c/prudential-lifeinsurance-assessment website, a site that organizes open competitions for data science and machine learning. The following is the details of the data that will be used in this study

- Data measuring 59381 x 128 in the .csv format.

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- There are 128 features that explain the attributes of life insurance applicants
- There are missing values in the dataset of 393110 from 13 different features in the dataset
- The dataset containing the missing values is shown in Fig 3.

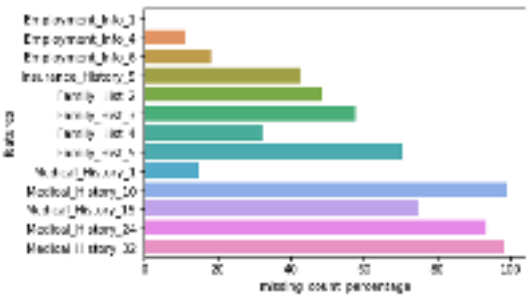

Fig 3. Percentage of features with missing values on Prudential data

Fig 3. shows the percentage of features with missing values from the least to the most. The results of the identification of the dataset against missing values are there are missing values on some features in the dataset. The lowest percentage of features with the least missing values is the Employment_Info_1 feature, and the highest percentage of features with the most missing values is the Medical_History_10 feature. The missing values result in a method that does not work optimally, even error, and causes the method accuracy to be low.

B. *Preprocessing Data*

The Prudential dataset contains categorical data that cannot be processed, namely in the form of A1, B1, C3, ... and so on in the Product_Info_2 feature. Therefore, a technique is needed to convert the data into a form that can be processed. In this study, a technique was used to overcome this problem, namely the encoder label technique.

The label encoder technique is used to convert non-numeric categorical data types to numerical data. The data in the Product_Info_2 feature is converted according to the sequence starting at 0,1,2,3,4, ..., 18. If the category in the Product_Info_2 feature reappears, it will be labeled according to the previous labeling

The non-numeric data set transformation becomes numeric for the Product_Info_2 feature can be seen in Figure 4.5. Data changes in the Product_Info_2 feature, namely A1 to 0, A2 to 1, A3 to 2, A4 to 3, A5 to 4, A6 to 5, A7 to 6, A8 to 7, B1 to 8, B2 to 9, C1 to 10, C2 becomes 11, C3 becomes 12, C3 becomes 13, D3 becomes 14, D2 becomes 15, D3 becomes 16, D4 becomes 17, and E1 becomes 18.

C. *Learning*

In the early stages of the simulation process, the data is divided into two parts randomly, namely training data and testing data with a percentage of 20% data testing and 80% training data. Then 80% of the training data is divided into two parts pickled with a proportion of 20% data testing and 80% training data. This new data sharing process is done five times, resulting in five new datasets.

Maximum depth	Accuracy
5	0.57659
10	0.59569
15	0.59659
20	0.59017
25	0.58354
30	0.57897
35	0.57715

Then, each dataset is carried out parameter optimization, which aims to determine the best parameter value of the model in the training data provided. If a parameter in XGBoost is optimized, the other hyperparameter values are the default value.

- The parameters used in building the XGBoost model are:
- The maximum depth which is the depth of the tree.
- Minimum child weight, which is the minimum amount of weight obtained from the derivative of both error (hessian) functions.
- Subsample, namely the comparison or subsampling ratio of the dataset proposed to build the CART.
- Colsample bytree, which is a comparison or subsampling dataset ratio used to construct each tree in a row and prevent overfitting.
- Learning rate, known as shrinkage techniques that work in each tree boosting and reduce the influence of each tree and leaf to improve the XGBoost model
- Alpha, namely L1 regularization and lambda, namely L2 regularization obtained from each iteration

D. *Evaluation*

The evaluation in this study is based on the value of quadratic weighted kappa, which measures the agreement (agreement) between two raters in classifying objects. The kappa value is between 0 and 1. The more the kappa value approaches 1, the better the model is formed. The response variable has eight possible ratings, and each application is characterized by tuples whose values correspond to those of Rater A (actual target) and Rater B (target prediction results). The Quadratic weighted kappa calculation phase is as follows:

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}}$$

$O_{i,j}$: application received with rank i to actual target and rank j o target prediction results

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$w_{i,j}$: weights are calculated based on the difference between assessors: $w_{i,j} = \frac{(i-j)^2}{(N-1)^2}$

E: rating expectations are calculated, assuming that there is no correlation between rating scores.

A. Result and Discussion

The evaluation results of the XGBoost model on testing data are in the form of accuracy with kappa units. The best parameters that have been optimized will be used for the final evaluation. The evaluation results of the XGBoost model based on the optimized parameters will be written in TABLE I.until TABLE V.

TABLE I. ACCURACY RESULTS FROM MAXIMUM DEPTH OPTIMIZATION

Based on TABLE I, the optimal value is obtained for the maximum depth parameter at value 15 with the highest accuracy of 0.59659. The optimal maximum depth parameter value will be used to optimize the minimum child weight parameters. Furthermore, optimization of the minimum child weight parameters is done. The results of the optimization of the parameters of the minimum child weight can be seen in TABLE II.

TABLE II. ACCURACY RESULTS FROM MINIMUM CHILD WEIGHT OPTIMIZATION

Minimum Child Weight	Accuracy
1	0.59817
5	0.59590
10	0.59722
15	0.59737

Based on Table II obtained the optimal value for the minimum child weight parameter at value 1 with the highest accuracy of 0.59817. The optimal value of maximum depth and minimum child weight parameters will be used to optimize learning rate parameters. Next, the learning rate parameter optimization is done; the results of the optimization of the rate parameters can be seen in TABLE III.

Parameter Value	Accuracy	
	Subsample	Colsample bytree
0.5	0.59087	0.58806
0.6	0.59133	0.59055
0.7	0.59466	0.59260
0.8	0.59522	0.59358
0.9	0.59571	0.59438
1	0.59808	0.59808

TABLE III. ACCURACY RESULTS FROM LEARNING RATE OPTIMIZATION

0.1	0.59808
1	0.47647
1.5	0.38208

Based on Table III obtained the optimal value for the learning rate parameter at a value of 0.1 with the highest accuracy of 0.59808. The maximum parameter value, minimum child weight, and optimal learning rate will be used to optimize the subsample parameters. Furthermore, optimization of subsample and colsample bytree parameters is done; the results of accuracy from the optimization of the subsample and colsample bytree parameters can be seen in TABLE IV.

TABLE IV. ACCURACY RESULTS FROM SUBSAMPLE AND COLSAMPLE BY TREE OPTIMIZATION

Based on TABLE IV obtained the optimal value for the colsample bytree parameter at value 1 with the highest accuracy of 0.59808. Colsample bytree parameter values cannot be extended anymore, because the maximum value is 1. The maximum depth parameter, minimum child weight, learning rate, subsample, and optimal colsample bytree will be used to optimize alpha parameters. Furthermore, alpha and lambda parameters optimization is done, the result of alpha and lambda parameters optimization can be seen in TABLE V.

TABLE V. ACCURACY RESULTS FROM ALPHA AND LAMBDA OPTIMIZATION

Parameter Value	Accuracy	
	Alpha	Lambda
0	0.59808	0.59566
0.25	0.59826	0.59871
0.5	0.59829	0.59638
0.75	0.59951	0.59835
1	0.60064	0.60064

Based on Table V obtained alpha parameter at value 1 with the highest accuracy of 0.60064 and the optimal value for lambda parameters at value 1 with the highest accuracy of 0.60064. After optimizing all parameters in the XGBoost model, parameter optimization based on optimal parameter values is used to evaluate the initial testing data, the most optimal parameter values are obtained. Furthermore, based on the value of the best parameters, the results of the optimization are evaluated on testing data. The results of the evaluation obtained the best accuracy of the XGBoost model with a measurement unit of the kappa coefficient of 0.60730.

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The performance of the XGBoost model will be compared to other machine learning models that have been used to predict the risk level of life insurance applicants. Machine learning models that have been developed to solve cases of classification, among them are decision tree and random forest [17]. The treatment used is filling the missing values with the mean value and removing the missing values..

TABLE VI. ACCURACY RESULTS DECISION TREE AND RANDOM FOREST USING A MEAN TECHNIQUE

Machine Learning Model	Accuracy
XGBoost	0.60730
Decision Tree	0.41678
Random Forest	0.47599

Based on Table VI, we see that the three models of data processing were carried out to deal with missing values with the mean technique, which is to fill the missing values with the average value. The accuracy of the three comparison models is determined based on the kappa coefficient value. The comparison model that has the highest accuracy is the random forest model with an accuracy of 0.47599. Then the comparison model that has the lowest accuracy is the decision tree model with an accuracy of 0.41678. When compared with the XGBoost model, the highest accuracy is the XGBoost model with an accuracy of 0.60730. This happened because of missing values in all three models treated with the mean technique, where the value of the missing values was filled with the average value of life insurance data used. If at a feature, the average value is zero, then this will affect accuracy, where the accuracy of the model is low.

TABLE VII. ACCURACY RESULTS DECISION TREE AND RANDOM FOREST USING DROP TECHNIQUE

Machine Learning Model	Accuracy
XGBoost	0,60730
Decision Tree	0.41286
Random Forest	0.47623

Accuracy on the three models based on Table VII produces accuracy that is lower than the accuracy of the XGBoost model. The treatment with the drop technique also produces accuracy that is close to the mean technique. The accuracy of the xgboost model is still superior when compared to the three comparison models. In the learning process, the xgboost model does not fill the missing values with a value, but the value of the missing values will still be used in the learning process with the default direction found in the sparsity aware algorithm for splitting finding.

V. CONCLUSION

Based on the results of the research and analysis that have been conducted, it can be concluded that the Extreme Gradient Boosting (XGBoost) model can be applied in the case of prediction of the applicant's risk level in life insurance which is a case of multiclass classification in the context of machine learning. The best parameters after being optimized used in this study are maximum depth = 15, minimum child weight = 1, subsample = 1, colsample bytree = 1, learning rate = 0.1; alpha = 1, lambda = 1. The accuracy of the XGBoost model by using the optimized parameters shows high accuracy of 0.60730. The dataset used is included in the big data with quite a lot of missing values, but this has proven to not affect the performance of the XGBoost model itself. In the research, the XGBoost model is considered capable of handling missing values well.

The performance of the XGBoost model is superior compared to the decision tree, random forest, and Bayesian ridge models when predicting the risk level of life insurance applicants. This is evidenced by the three comparative models of learning with the treatment of the missing values, namely using the mean and drop techniques. But this does not change the accuracy of the three models into high-value accuracy.

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